

# Udayan Atreya

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## EDUCATION

### Master of Science, Computer Science

May 2026

San Jose State University, San Jose, US

Coursework: Applied Deep Learning, Retrieval Augmented Generation (RAG), Biometric Security with AI.

### Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures and Algorithms, Relational Database Management Systems, Object-Oriented Design.

## TECHNICAL SKILLS

**Web & Languages:** HTML, CSS, JavaScript, Python, Java, C#, SQL

**Frameworks & APIs:** FastAPI, .NET, REST API Integration, LangChain, RAG, HuggingFace, TensorFlow, PyTorch

**Cloud & Database:** AWS (API Gateway, Lambda, S3, DynamoDB), CI/CD, MongoDB, Pinecone, ChromaDB

**LLMs & AI Tools:** Claude, GPT-4o, Gemini, Ollama, Llama, DeepSeek, Groq, RAGAS

## PROJECT EXPERIENCE

### Agentic BMC Observability Platform (*FastAPI, LangChain, Gemini 1.5, MongoDB, AWS S3*)

[\[GitHub Link\]](#)

Jun 2025 – Jul 2025

- Built a responsive web observability platform using **FastAPI** with live **Grafana dashboards**, exposing clean REST API endpoints to exchange telemetry data between backend services and the web UI, achieving **95% query classification accuracy** and **30% reduction in manual triage time**.
- Engineered a frontend-to-backend REST API integration layer connecting a hybrid MongoDB + AWS S3 retrieval engine to web clients, **cutting data retrieval time by 75%** (2m to 30s) through well-documented, maintainable API contracts.
- Secured agentic hardware interventions via role-based access controls and human-in-the-loop validations, **preventing 100% of unauthorized state changes** with strict, auditable policy enforcement.

### Evaluator-Optimizer RAG (*LangChain, GPT-4o, Gemini 1.5 Pro, Llama-3, ChromaDB, Whisper*)

[\[GitHub Link\]](#)

Aug 2025 – Mar 2026

- Architected an agentic RAG framework with an Evaluator-Optimizer feedback loop, **reducing hallucination by 25%** and **improving answer accuracy by 18%** over baselines for Language-Based Video Reasoning systems.
- Engineered a REST API-accessible evaluation suite using RAGAS and Llama-3-70b (LLM-as-a-judge) across 500+ STEM QA pairs, quantifying Faithfulness, Answer Relevancy, Context Precision, Context Recall, and Semantic Similarity.
- Optimized the latency-accuracy trade-off via Dynamic Chunking and MMR search, **reducing context redundancy by 40%** while sustaining **90%+ semantic coverage** for scalable, web-accessible AI applications.

## WORK EXPERIENCE

### Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

### Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Architected a user-facing notification system backed by **AWS REST APIs (API Gateway, Lambda, DynamoDB)**, delivering real-time remediation stage updates to web clients and driving a **20% increase in DAU** post-launch.
- Optimized backend REST API performance by redesigning **DynamoDB access patterns** and CRUD microservices, **reducing frontend-consumed retrieval latency by 37%** and ensuring responsive, high-performing web experiences.
- Built a **ChatOps web platform** routing natural language queries to backend automation workflows via REST API integration, reducing operational overhead by **40%** (~5m to ~1m) for cross-functional teams.
- Delivered a **code vulnerability remediation** web solution using serverless AWS (Lambda, S3, EventBridge) and Azure DevOps CI/CD, providing clean, documented REST API integrations safeguarding **400+ applications** across organizations.
- Developed an analytics-driven recommendation engine as a user-facing web feature, implementing weighted scoring logic to personalize tool suggestions across 100+ automation assets, **reducing discovery time by 30%**.
- Designed a single-touch server patching solution integrating ServiceNow alerts with a serverless AWS backend, reducing Mean Time to Resolution (**MTTR**) by **80%** through automated backend-to-frontend status synchronization.